EV MARKET ANALYTICS

Project Report: Data Analytics and Visualisation using Tableau

Prepared by: Prisha Jain

Date: 31 October 2026

1. Introduction

1.1 Project Overview

EV Market Analytics is an individual data analytics project focused on exploring the Electric Vehicle (EV) market. The project studies vehicle specifications, manufacturers, market performance, battery technology, pricing, customer ratings and environmental impact. The dataset contains 37 attributes, and Tableau Public is used to create interactive visualisations and a data story. A Flask-based web application is also developed to provide access to the Tableau dashboard and story.

1.2 Purpose

The purpose of this project is to organise EV market information and present it in a visual format. It helps users explore different aspects of electric vehicles, compare manufacturers and models, and understand market and vehicle-performance information more easily.

2. Ideation Phase

2.1 Problem Statement

The electric vehicle market includes many manufacturers, models, prices and technical specifications. When this information is available only in raw tabular form, it can be difficult to compare vehicles and identify useful patterns. This project aims to make EV market data easier to understand through interactive Tableau visualisations and a web-based platform.

2.2 Brainstorming

The project focuses on the following analytical areas:

- EV manufacturer and vehicle-model comparison.
- Regional EV market distribution and annual vehicle sales.
- Battery capacity, battery type and driving range.
- Charging performance and energy efficiency.
- Vehicle pricing and ownership-related costs.
- Customer ratings, safety and environmental impact.

2.3 Proposed Direction

The selected solution consists of an interactive Tableau Dashboard, a Tableau Story that presents the analysis in a sequence, and a Flask web application that embeds both Tableau resources.

3. Requirement Analysis

3.1 User Requirements

The project is designed for students, researchers, analysts and people interested in electric mobility. Users should be able to explore EV market information, compare vehicle characteristics and view the analysis without manually examining the complete dataset.

3.2 Functional Requirements

- Display EV market information through visualisations.
- Support comparisons between manufacturers and vehicle models.
- Explore battery capacity, driving range and charging performance.
- Review vehicle pricing and annual sales information.
- Examine customer ratings, safety and environmental impact.
- Provide access to the Tableau Dashboard and Tableau Story.
- Integrate Tableau content into a Flask-based web application.

3.3 Non-functional Requirements

- Clear and readable visualisations.
- Simple navigation and user-friendly interface.
- Accessible Tableau content through Tableau Public.
- Consistent presentation across the web application.

3.4 Dataset Description

The project uses an Electric Vehicle dataset containing 37 columns. The dataset includes vehicle identification and model information, manufacturer and manufacturing country, model year, market region, vehicle category, battery capacity and type, battery life and degradation, driving range, energy efficiency, motor power, charging capacity and time, vehicle price, insurance cost, warranty, resale value, safety ratings, autonomous driving level, annual sales, customer ratings and environmental impact.

3.5 Technology Stack

Technology	Purpose
Python	Backend programming
Flask	Web application framework
HTML5	Webpage structure
CSS3	Website styling
JavaScript	Frontend interactivity
Tableau Public	Data visualisation and storytelling
Visual Studio Code	Development environment

4. Project Design

4.1 Problem Solution Fit

The project addresses the difficulty of understanding a large EV dataset in raw tabular form. Tableau visualisations make the information easier to explore, while the story format helps present the analysis in an organised sequence.

4.2 Proposed Solution

- Tableau Dashboard: presents EV market information through interactive visualisations.
- Tableau Story: arranges the project's visual analysis in a structured storytelling format.
- Flask Web Application: provides a homepage, project overview and embedded Tableau Dashboard and Story.

4.3 Solution Architecture

EV dataset → Tableau data visualisation and story → Tableau Public publishing → Flask web application with embedded Tableau content → End user.

4.4 Web Application

The Flask-based application introduces EV Market Analytics and provides a user-friendly interface for exploring the project. It includes a homepage, project overview, embedded Tableau Dashboard and embedded Tableau Story.

5. Project Planning and Scheduling

5.1 Project Planning

Phase	Activities	Output
1. Understanding	Identify the project problem and objectives.	Project scope
2. Dataset review	Review the EV dataset and identify relevant attributes.	Dataset understanding
3. Visualisation	Create visualisations using Tableau.	Tableau visualisations
4. Dashboard and Story	Arrange visualisations into a dashboard and story.	Dashboard and Story
5. Web integration	Build the Flask interface and embed Tableau content.	Web application
6. Documentation	Prepare project documentation and links.	Final report

6. Functional and Performance Testing

6.1 Testing

The following checks can be used to verify the project:

Test	Expected result
Tableau Dashboard link	Dashboard opens correctly on Tableau Public.
Tableau Story link	Story points are accessible.
Dashboard interaction	Available filters and chart interactions respond correctly.
Flask homepage	Homepage and project overview display correctly.
Embedded Tableau content	Dashboard and Story are visible within the web application.
Repository access	Project files and documentation are accessible on GitHub.

6.2 Expected Result

The expected result is an interactive visual analysis of EV market information, a Tableau Story that presents the analysis in sequence, and a Flask web application that provides access to both Tableau resources.

7. Results

7.1 Project Output

The project provides an interactive Tableau Dashboard and a Tableau Story published on Tableau Public. The dashboard allows users to explore the EV market through the available visualisations. The story presents the analysis in a structured sequence. The Flask web application provides an additional interface to access these resources.

7.2 Key Analytical Areas

- EV manufacturer and model comparisons.
- Regional EV market distribution.
- Annual vehicle sales.
- Battery capacity and driving range.
- Charging performance.
- Vehicle pricing and ownership costs.
- Customer ratings and safety.
- Environmental impact of electric vehicles.

Note: Exact numerical findings and chart-specific conclusions should be taken from the published Tableau workbook. They have not been added here without confirmation from the live visualisations.

8. Advantages and Disadvantages

Advantages

- Makes a large EV dataset easier to explore.
- Presents several aspects of the EV market in visual form.
- Provides a structured Tableau Story for presenting the analysis.
- Offers a web interface through Flask.
- Makes the project accessible through Tableau Public and GitHub.

Disadvantages

- The analysis depends on the quality and completeness of the dataset.
- The dashboard reflects the data available in the connected dataset and may not show real-time market updates.
- The accuracy of comparisons depends on consistent data values and categories.

9. Conclusion

EV Market Analytics demonstrates how data visualisation can make electric vehicle market information easier to understand and compare. The project uses Tableau Public to present the analysis through a Dashboard and Story, and a Flask web application to provide access to these resources. It covers important areas such as manufacturers, vehicle specifications, battery technology, pricing, sales and environmental impact.

10. Future Scope

- Integrate regularly updated EV market datasets.
- Add forecasting of EV sales and market trends.
- Include more detailed manufacturer and model comparisons.
- Add further interactive filters and visualisations.
- Integrate a database for improved data management.
- Deploy the Flask application on a production hosting platform.
- Expand the analysis with additional EV market indicators.

11. Appendix

11.1 Project Information

Detail	Information
Project Name	EV Market Analytics
Prepared by	Prisha Jain
Project Type	Individual Project
Domain	Data Analytics and Data Visualisation
Technology	Python, Flask, Tableau Public, HTML, CSS and JavaScript

11.2 Project Links

GitHub Repository: https://github.com/en25cs3010712/EV-Market-Analytics

Tableau Dashboard: Open Dashboard

Tableau Story: Open Story

11.3 Project Structure

```
EV-Market-Analytics/
■■■ app.py
■■■ templates/index.html
```

```
■■■ static/css/style.css
■■■ static/js/main.js
■■■ Dataset/
■■■ Documentation/
■■■ Tableau/
■■■ Demo/
■■■ README.md
```